

Mechanical Engineering Experiments — Report 2

機械工程實驗(一) 第二次報告

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Q1:

- (a) . **Plot** a strain to $V_{average}$ plot, with strain at the x-axis and $V_{average}$ at the y-axis. Also mark your STD range on each data point. ** Follow the preview question Q4 for latter week unit to estimate your gauge strain.

Table 1. Experiment recordings for strain gauge experiment

	$V_{initial}$	$V_{reading\ 1}$	$V_{reading\ 2}$	$V_{reading\ 3}$	$V_{average}$	V_{STD}
Curvature 1 (550mm)	-0.010 V	0.28 V	0.28 V	0.27 V	0.28 V	0.006 V
Curvature 2 (750mm)	-0.006 V	0.20 V	0.19 V	0.19 V	0.19 V	0.006 V
Curvature 3 (950mm)	-0.007 V	0.16 V	0.16 V	0.16 V	0.16 V	0 V
	$V_{initial}$	$V_{reading}$				
Shunt Calibration	0.007 V	-0.840 V	Strip thickness	0.63mm		

Gauge strain = $\varepsilon_x = \frac{y}{r}$ y: distance away from neutral axis r: curvature radius

$$\text{Curvature 1 (550mm): } gauge\ strain = \frac{\frac{0.63}{2}}{550 + \frac{0.63}{2}} \approx 0.0005727$$

$$\text{Curvature 2 (750mm): } gauge\ strain = \frac{\frac{0.63}{2}}{750 + \frac{0.63}{2}} = 0.00042$$

$$\text{Curvature 3 (950mm): } gauge\ strain = \frac{\frac{0.63}{2}}{950 + \frac{0.63}{2}} \approx 0.0003316$$

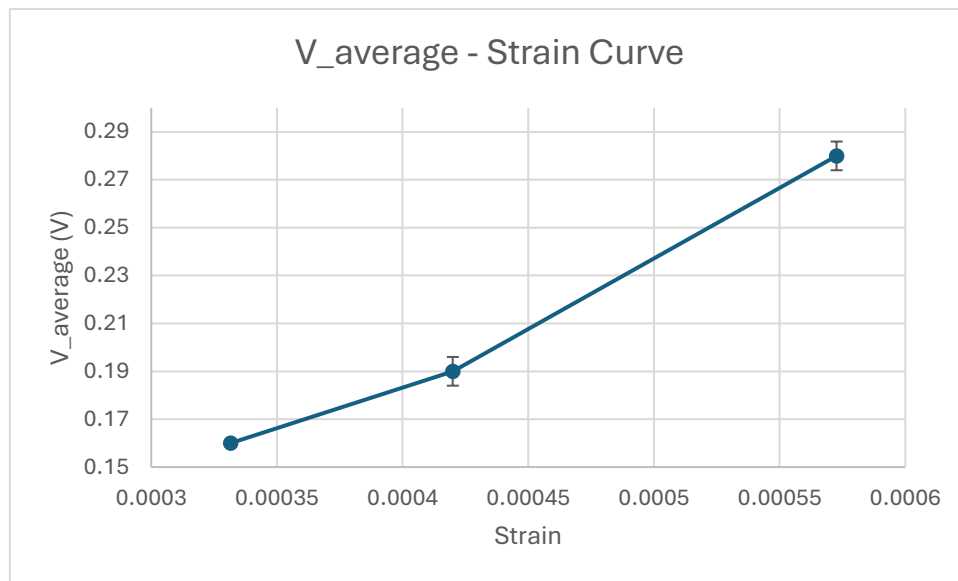


Figure 1. V_{average} — Strain curve

- (b) . Apply a linear regression for your result of (a). This demo video may help you to complete a linear regression by applying the Excel. How to do a linear regression on excel - YouTube ** Report your linear regression formula and the R^2 value.

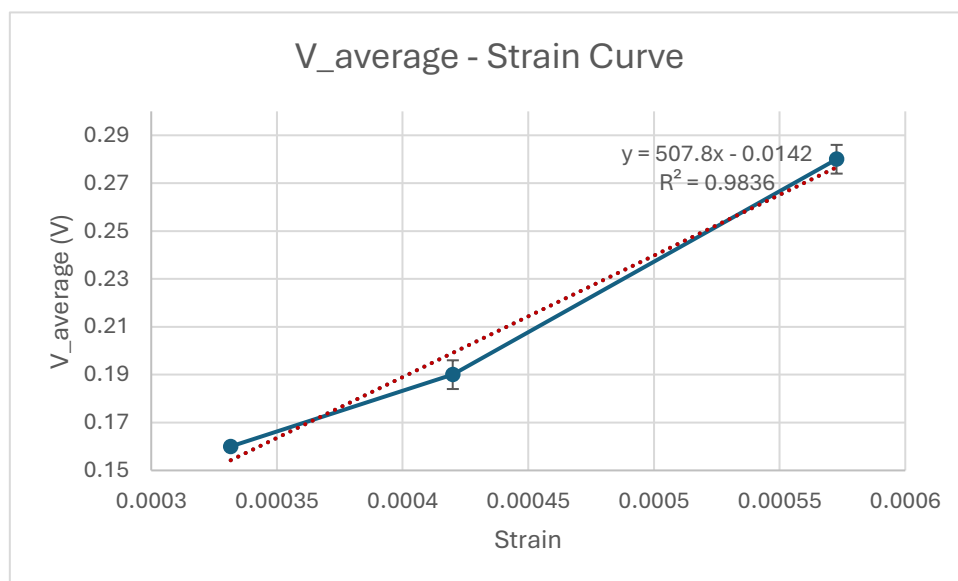


Figure 2. V_{average} — Strain curve (with linear regression)

$$V_{\text{average}} = 507.8 \times \text{Strain} - 0.0142$$

$$R^2 = 0.9836$$

(c) . Next, we had obtained a shunt calibration voltage reading. Apply this value to your linear regression to obtain a virtual strain $\epsilon_{virtual}$. We can then evaluate an experimental gauge factor S_g by the following equation: $S_g = \left| \frac{\frac{\Delta R}{R}}{\epsilon_{virtual}} \right|$.

Calculate your experimental gauge factor and its error with respect to the gauge factor provided by manufacturer (2.12). Involved parameters are defined below:
R: Resistance of your strain gauge (If you forgot to record this value, apply 362Ω).

$\Delta R = R - R \parallel R_{shunt}$: Resistance variation after a $100\text{ k}\Omega$ shunt resistance (R_{shunt}) is parallel to your strain gauge. **Hint: recall from p.9, how do we estimate an equivalent resistance ($R \parallel R_{shunt}$) in a parallel circuit?

$$V_{average} = 507.8 \times Strain - 0.0142$$

$$V_{shunt} = 507.8 \times \epsilon_{virtual} - 0.0142$$

$$V_{shunt} = -0.840V$$

$$\epsilon_{virtual} \approx -0.001626$$

$$\Delta R = R - R \parallel R_{shunt} = R - \frac{R \times R_{shunt}}{R + R_{shunt}} = 362 - \frac{362 \times 100000}{362 + 100000} \approx 1.3057\Omega$$

$$S_g = \left| \frac{\frac{\Delta R}{R}}{\epsilon_{virtual}} \right| = \left| \frac{\frac{1.3057}{362}}{-0.001626} \right| \approx 2.2183$$

$$Error\ Percentage = \frac{S_g - S_{g,manufacturer}}{S_{g,manufacturer}} = 4.637\%$$

Q2:

(a) . **Plot** a weight to $V_{average}$ plot, with weight at the x-axis and $V_{average}$ at the y-axis. Also mark your STD range on each data point. Please plot both the loading curve and the unloading curve on the same plot. ** Hint:

$$V_{1kg,loading,average} = \frac{V_{1kg,loading1} + V_{1kg,loading2} + V_{1kg,loading3}}{3} \text{ with its}$$

corresponding STD...and so on.

Table2. Experiment recordings for load cell experiment

	$V_{initial}$	V_{1kg}	V_{2kg}	V_{3kg}	V_{4kg}	V_{5kg}
Loading 1	1 mV	10.6 mV	19.6 mV	28.7 mV	37.8 mV	47 mV
Unloading 1	1 mV	10.2 mV	19.4 mV	28.7 mV	37.9 mV	
Loading 2	1 mV	10.2 mV	19.5 mV	28.7 mV	37.9 mV	47 mV
Unloading 2	1.4 mV	10.7 mV	19.8 mV	28.9 mV	38 mV	
Loading 3	1.4 mV	11 mV	20.1 mV	29.4 mV	38.8 mV	48 mV
Unloading 3	2 mV	11.2 mV	20.3 mV	29.5 mV	38.8 mV	

$$V_{initial,loading,average} \approx 1.1333 \text{ mV}$$

$$V_{initial,loading,STD} \approx 0.2309 \text{ mV}$$

$$V_{initial,unloading,average} \approx 1.4667 \text{ mV}$$

$$V_{initial,unloading,STD} \approx 0.5033 \text{ mV}$$

$$V_{1kg,loading,average} \approx 10.6 \text{ mV}$$

$$V_{1kg,loading,STD} \approx 0.4 \text{ mV}$$

$$V_{1kg,unloading,average} \approx 10.7 \text{ mV}$$

$$V_{1kg,unloading,STD} \approx 0.5 \text{ mV}$$

$$V_{2kg,loading,average} \approx 19.7333 \text{ mV}$$

$$V_{2kg,loading,STD} \approx 0.3215 \text{ mV}$$

$$V_{2kg,unloading,average} \approx 19.6667 \text{ mV}$$

$$V_{2kg,unloading,STD} \approx 0.3786 \text{ mV}$$

$$V_{3kg,loading,average} \approx 28.9333 \text{ mV}$$

$$V_{3kg,loading,STD} \approx 0.4041 \text{ mV}$$

$$V_{3kg,unloading,average} \approx 29.0333 \text{ mV}$$

$$V_{3kg,unloading,STD} \approx 0.4163 \text{ mV}$$

$$V_{4kg,loading,average} \approx 38.1667 \text{ mV}$$

$$V_{4kg,loading,STD} \approx 0.5508 \text{ mV}$$

$$V_{4kg,unloading,average} \approx 38.2333 \text{ mV}$$

$$V_{4kg,unloading,STD} \approx 0.4933 \text{ mV}$$

$$V_{5kg,loading,average} \approx 47.3333 \text{ mV}$$

$$V_{5kg,loading,STD} \approx 0.5774 \text{ mV}$$

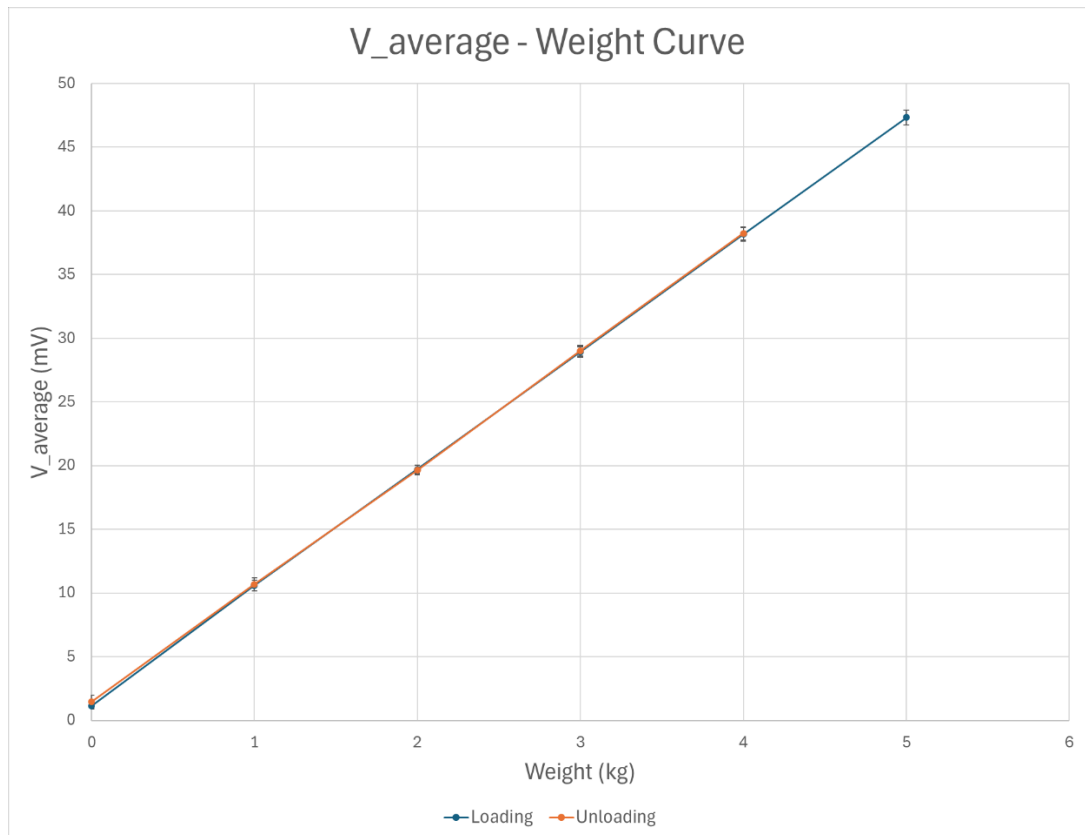


Figure 3. $V_{average}$ — Weight curve

(b) . **(Research by yourself)** Refer to the following plot, you can understand the definition of linearity and reproducibility; **define a criterion** to judge how well is the linearity and reproducibility of your load cell? **** Note:** “it looks like linear and matches with each other well” is not a criterion. A criterion requires a mathematical description.

Linearity:

If a system or function is linear, the output (Voltage) changes proportionally with the input (weight). It will form a straight line through the origin, and can be expressed by:

$$V = a \times Weight + b \quad (b \approx 0) \quad \rightarrow \quad V = a \times Weight$$

a: constant representing the slope of the equation.

b: constant representing the y-intercept of the equation. Often ≈ 0 to satisfy the linear property.

Linearity Criterion:

We can define our linearity criterion by setting up a threshold of the R^2 value for the linear regression. (R^2 value: Coefficient of Determination. Will $\rightarrow 1$ if the original data points are perfectly linear.)

For example, the linearity criterion can be defined by the threshold $R^2 \geq 0.95$ to determine whether it has a good linear fit or not.

For our experiment data:

$$\text{Loading: } V = 9.2257 \times \text{Weight} + 1.2524 \quad R^2 \approx 1$$

$$\text{Unloading: } V = 9.1867 \times \text{Weight} + 1.4467 \quad R^2 \approx 1$$

Both R^2 values exceed our threshold of 0.95, showing that the two set of data points have an excellent linear fit.

Reproducibility:

Reproducibility is the ability of an experiment to produce consistent results when it is repeatedly performed. It reflects the robustness and reliability of a experiment.

Reproducibility Criterion:

If repeated measurements $V_{\text{loading},i}$ and $V_{\text{unloading},i}$ are taken, reproducibility can be quantified by calculating the Root Mean Square Error between $V_{\text{loading},i}$ and $V_{\text{unloading},i}$, which can be defined as:

$$\text{Root Mean Square Error} = \sqrt{\frac{\sum_{i=1}^n (V_{\text{loading},i} - V_{\text{unloading},i})^2}{n}}$$

A small Root Mean Square Error means high reproducibility.

Then, we can define our Root Mean Square Error threshold (for example:

$\text{Root Mean Square Error} \leq 0.5 \text{ mV}$) to determine whether it has a good reproducibility or not.

For our data:

$$\text{Root Mean Square Error} = \sqrt{\frac{\sum_{i=1}^n (V_{\text{loading},i} - V_{\text{unloading},i})^2}{n}} \approx 0.1673 \text{ mV},$$

which is smaller than the threshold of 0.5mV, showing that the experiment has excellent reproducibility.

Q3:

By your data in Table 2, choose one arbitrary loading curve and explain if you forgot to “zero” your load cell, how does it affect your true loading curve?

If forgot to zero load cell before experiment, the load cell will have a non-zero initial voltage before loading starts. Then every measured point will include that offset: $V_{measure} = V_{true} + V_{offset}$.

The shape of the curve does not change because the proportional relationship between weight and voltage is still valid. However, the entire curve shifts vertically by the offset amount V_{offset} .

Choose loading 1 for example:

If we forgot to zero it before loading (*e.g.*, $V_{initial} = 10 \text{ mV}$), every subsequent voltage reading will have a 10mV offset:

$V_{measure} = V_{true} + 10 \text{ mV}$. The curve will shift from the blue one to the orange one in figure 4, and all measurements will reflect a higher value than the actual load-induced voltages

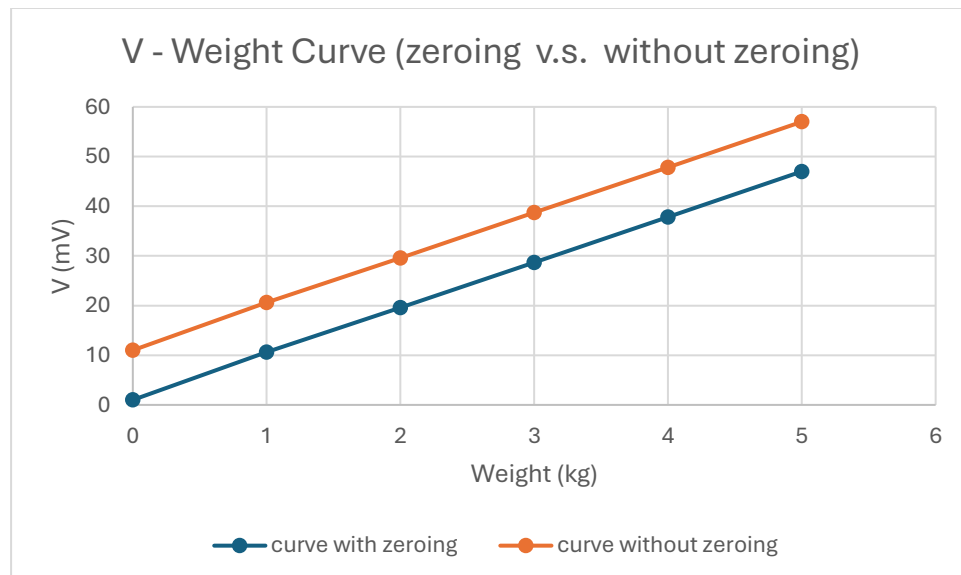


Figure 4. V – Weight curve (zeroing v.s. without zeroing)